



Sanger Sequencing of glucocerebrosidase (GBA) in PPMI

Summary

Sanger sequencing was performed on whole-blood extracted DNA samples collected according to the Parkinson's Progression Markers Initiative Research Biomarkers Laboratory Manual. Exons 1-11 within the glucocerebrosidase gene were Sanger sequenced and screened for variants E365K, A495P, K13R, L444P, N370S, R78C, R83C, T336S and T408M in 612 PPMI subjects.

Method

All exons within the glucocerebrosidase gene were selectively amplified in three fragments ranging from 1.7kb to 3kb in length. A fragment encompassing exons 1–5 was amplified using the forward primer 5'-CCTAAAGTTGT- CACCCATAC-3' and the reverse primer 5'- AGCAGACCTACCCTACAGTTT-3' (annealing temperature 57 C, extension time 3 min). A second fragment covering exons 5–7 was amplified using the forward primer 5'-GACCTCAAATGATATACCTG-3' and the reverse primer 5'-AGTTTGGGAGCCAGTCATTT-3' (annealing temperature 58.5 C, extension time 2 min). Finally, a fragment extending across exons 8–11 was amplified using the forward primer 5'- TGTGTGCAAGGTCCAGGATCAG-3' and the reverse primer 5'-ACCACCTAGAGGGGAAAGTG-3' (annealing temperature 61 C, extension time 90 sec).

Amplified segments were purified prior to sequencing using QIAquick PCR Purification Kit (Qiagen, Santa Clarita, CA). Cycle sequencing was accomplished using Big Dye Terminator Cycle Sequencing kit (Applied Biosystems, Porter City, IA) and fragments were sequenced on Applied Biosystems 3730 Genetic Analyzer using both forward and reverse primers. Sequences were analyzed using Sequencer Software 5.2.4 (Gene Codes Corp).





References

Parkinson's Progression Marker Initiative Research Biomarkers Laboratory Manual (Biologic Manual) <http://www.ppmi-info.org/wp-content/uploads/2011/05/PPMI-Biologics-Manual-April-2011-FINAL.pdf>.

Tayebi N, Cushner S, Sidransky E. 1996a. Differentiation of the glucocerebrosidase gene from pseudogene by long-template PCR: Implications for Gaucher disease. *Am J Hum Genet* 59:740–741.

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